CS1010 Discrete Mathematics for Computer Science

Shibam Ghosh

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Department of CSE, IIT Hyderabad



Conditional Probability

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Definition

Let E and F be events with $\Pr[F] > 0$. The conditional probability of E given F, denoted by $\Pr[E|F]$, is defined as:

$$\Pr[E|F] = \frac{\Pr[E \cap F]}{\Pr[F]}$$

$\mathcal{Q}$. A card is drawn at random from a standard deck of 52 cards. Suppose you are told that the card drawn is a **heart**. What is the probability that it is a **face card**?

$\mathcal{A}$. Let $E = \{\text{face card}\}$, $F = \{\text{heart}\}$. Since there are 13 hearts and only 3 of them are face cards (Jack, Queen, and King),

$$\Pr(E \mid F) = \frac{\Pr(E \cap F)}{\Pr(F)} = \frac{3/52}{13/52} = \frac{3}{13}.$$

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Conditional Probability Example

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$\mathcal{A}$. Let E be the event of at least two consecutive 0s, and F be the event the first bit is 0.

$$E \cap F = \{0000, 0001, 0010, 0011, 0100\} \implies p(E \cap F) = 5/16$$

Since 8 bit strings start with 0, $p(F) = 8/16 = 1/2$. Thus:

$$\Pr[E|F] = \frac{5/16}{1/2} = \frac{5}{8}$$



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$\mathcal{Q}$. Are **two boys** and **at least one boy** independent for a two-child family?

$\mathcal{A}$. We have $\Pr[E] = 1/4$, $\Pr[F] = 3/4$, and $\Pr[E \cap F] = 1/4$.

$$\Pr[E] \Pr[F] = \frac{3}{16} \neq \frac{1}{4} = \Pr[E \cap F]$$

Therefore, E and F are **not independent**.

Pairwise and Mutual Independence

Definition

The events $E_1, E_2, \ldots, E_n$ are **pairwise independent** if and only if

$$\Pr(E_i \cap E_j) = \Pr(E_i)\Pr(E_j) \text{ for all } i < j \leq n.$$

They are **mutually independent** if

$$\Pr\left(\bigcap_{j=1}^{m} E_{i_j}\right) = \prod_{j=1}^{m} \Pr(E_{i_j})$$

whenever i_j are integers with $1 \leq i_1 < i_2 < \cdots < i_m \leq n$ and $m \geq 2$.

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$$F = S \cap F = (E \cup \overline{E}) \cap F = (E \cap F) \cup (\overline{E} \cap F).$$

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$$\Pr(\overline{E} \cap F) = \Pr(F) - \Pr(E \cap F) = \Pr(F) - \Pr(E)\Pr(F) = \Pr(F)[1 - \Pr(E)].$$

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$\mathcal{A}$. Thus,

$$\boxed{\Pr(\overline{E} \cap F) = \Pr(\overline{E})\Pr(F)}.$$

Therefore, $\overline{E}$ and F are independent.



Q&A: Independence

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$\mathcal{Q}$. $\Pr(i\text{th child is a boy}) = 0.51 - \frac{i}{100}$.

$\mathcal{A}$. $\Pr(i\text{th child is a girl}) = 0.49 + \frac{i}{100}$. By independence,
$\Pr(\text{no boys}) = \prod_{i=1}^{5} \left(0.49 + \frac{i}{100}\right)$.
Thus, $\Pr(\text{no boys}) = (0.50)(0.51)(0.52)(0.53)(0.54) \approx \boxed{0.0382}$.

Random Variables

Random Variables

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- A **random variable** is a function from the sample space of an experiment to the set of real numbers.
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Roll two fair six-sided dice. Let $X(d_1, d_2)$ be the random variable denoting the **sum** of the two numbers. For example,

- $X(1,1) = 2$
- $X(2,5) = 7$
- $X(6,6) = 12$

Distribution: $P(X = 2) = \frac{1}{36}, \quad P(X = 7) = \frac{6}{36}, \quad P(X = 12) = \frac{1}{36}.$



Random Variables Example

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$\mathcal{A}$.
- $X(HHH) = 3$, $X(TTT) = 0$
- $X(HHT) = X(HTH) = X(THH) = 2$
- $X(TTH) = X(THT) = X(HTT) = 1$

Distribution:

$\Pr(X = 3) = 1/8, \Pr(X = 2) = 3/8, \Pr(X = 1) = 3/8, \Pr(X = 0) = 1/8.$

Probability Distribution of a Random Variable

For a random variable X, define

$$f(x) = \Pr(X = x).$$

The function f is called the **probability distribution** of X.

For a discrete random variable,

$$f(x) \geq 0, \qquad \sum_{x \in X(S)} f(x) = 1.$$

For example, if X is the number of heads in three fair coin tosses,

$$\Pr(X = 0) = \frac{1}{8}, \quad \Pr(X = 1) = \frac{3}{8}, \quad \Pr(X = 2) = \frac{3}{8}, \quad \Pr(X = 3) = \frac{1}{8}.$$



Discrete and Continuous Random Variables

Discrete random variable

A random variable X is discrete if it takes values in a finite or countable set.

Its probability mass function is

$$f_X(x) = \Pr(X = x).$$

Continuous random variable

A random variable X is continuous if there exists a non-negative function f such that, for every interval B,

$$\Pr(X \in B) = \int_B f(x)\, dx.$$

The function f is called the **density function**.



Cumulative Distribution Function

For a random variable X, define

$$F(x) = \Pr(X \leq x).$$

This is called the **cumulative distribution function (CDF)**.

Equivalently,

$$F(x) = \Pr\big(\{\omega \in \Omega : X(\omega) \leq x\}\big).$$

For a continuous random variable with density f,

$$F(x) = \int_{-\infty}^{x} f(t)\, dt.$$



Random Variables

Some Useful Distributions

Bernoulli Trials

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Suppose an experiment can have only two possible outcomes (e.g., coin flipping). Each such performance is a **Bernoulli trial**: One outcome is a *success* (p) and the other a *failure* (q), where $p + q = 1$.



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$\mathcal{Q}$. A coin is biased so that $\Pr(H) = 2/3$. What is the probability of exactly four heads in seven flips?

$\mathcal{A}$. The number of ways 4 of the 7 flips can be heads is $C(7, 4)$. The probability of each such outcome is $(2/3)^4(1/3)^3$.

$$\text{Probability} = C(7,4)\left(\frac{2}{3}\right)^4\left(\frac{1}{3}\right)^3 = \frac{35 \cdot 16}{2187} = \frac{560}{2187}$$

k Successes in n Independent Bernoulli Trials

Theorem

The probability of exactly k successes in n independent Bernoulli trials is:

$$b(k : n, p) = C(n, k)p^k q^{n-k}.$$



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Proof: The probability of any specific outcome with k successes and $n - k$ failures is $p^k q^{n-k}$. Since there are $C(n, k)$ such n-tuples, the total probability is

$$b(k : n, p) = C(n, k)p^k q^{n-k}$$

Q&A: Bernoulli Trials

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$\mathcal{A}$. There are either 0 or 1 failures: $\Pr(\text{at most one failure}) = p^n + np^{n-1}(1-p)$.

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$\mathcal{Q}$. The probability of at least two failures

$\mathcal{A}$. Use the complement of at most one failure:
$\Pr(\text{at least two failures}) = 1 - p^n - np^{n-1}(1-p)$.



Bernoulli Random Variable

A **Bernoulli random variable** models the outcome of a single experiment with two possible outcomes.

$$X = \begin{cases} 1, & \text{success}, \\ 0, & \text{failure}. \end{cases}$$

If the probability of success is p, then

$$\Pr(X = 1) = p, \qquad \Pr(X = 0) = 1 - p.$$

- A manufactured component is defective or non-defective.
- A password guess is correct or incorrect.
- A server request succeeds or fails.
- A coin toss: Head or Tail.



From Bernoulli to Binomial

Suppose we perform the same Bernoulli experiment n times, independently, with success probability p. Let X be the **number of successes**. Then

$$X \sim \text{Binomial}(n, p).$$

Its probability distribution is

$$\Pr(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \text{ for } k = 0, 1, \ldots, n.$$

The important question is:

How many successes occur in a fixed number of trials?



Bernoulli: What happens in one trial? **Binomial:** How many successes?

Applications of the Binomial Distribution

The binomial distribution is useful whenever:

- there is a fixed number n of trials,
- each trial has two possible outcomes,
- the probability of success is p,
- the trials are independent.

Examples:

- Number of defective components in a sample of 100.
- Number of correct answers when guessing on 10 true/false questions.
- Number of heads obtained in n tosses of a biased coin.



Geometric Random Variable

Let X denote the number of trials needed to obtain the **first success**, where each trial has success probability p. Then

$$\Pr(X = k) = (1 - p)^{k-1} p, \qquad k = 1, 2, \ldots$$

Since exactly $k - 1$ failures are followed by one success.

The probabilities sum to one:

$$\sum_{k=1}^{\infty} (1 - p)^{k-1} p = 1.$$



Applications of the Geometric Distribution

The geometric distribution answers the question:

How many trials do we need until the first success?

Examples:

- Number of random guesses until the correct password is found.
- Number of trials until a defective component is detected.
- Number of coin tosses until the first head appears.

If the success probability in each trial is p, then

$$\Pr(X = k) = (1 - p)^{k-1} p.$$



Geometric Distribution: Waiting for Success

Let X be the number of trials until the first success. The probability that we need *more than k trials* is

$$\Pr(X > k) = (1 - p)^k.$$

Why?

$$X > k \iff \text{the first } k \text{ trials are all failures.}$$

Therefore,

$$\boxed{\Pr(X > k) = (1 - p)^k}.$$

For example, if $p = 0.2$, $\Pr(X > 5) = (0.8)^5$. This is often easier to calculate than summing $\Pr(X = 6) + \Pr(X = 7) + \cdots$.



Which Distribution?

Suppose a random experiment involves success/failure trials.

Question	Distribution
Outcome of one trial	Bernoulli
Number of successes in n trials	Binomial
Number of trials until first success	Geometric



Poisson Distribution

The Poisson distribution models the number of times an event occurs during a fixed interval of time, distance, area, etc.

Examples:

- Number of calls received by a call center in one minute.
- Number of defects in one meter of manufactured cable.
- Number of α-particles detected in one second.
- Number of requests received by a server in one minute.

If X is Poisson with parameter $\lambda > 0$, then

$$\boxed{\Pr(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}} \text{ where } k = 0, 1, 2, \ldots.$$



Interpreting the Poisson Parameter

For a Poisson process, λ represents the **average rate of occurrence** over the interval being considered. For example, suppose a server receives

3.2 requests per second.

- For a one-second interval, we use $\lambda = 3.2$.
- For a two-second interval, $\lambda = 6.4$.
- For a 10-second interval, $\lambda = 32$.

Thus, the parameter changes when the size of the interval changes.



Application of the Poisson Distribution

Suppose a server receives

$$3.2 \text{ requests per second.}$$

Let X denote the number of requests in one second.

$\mathcal{Q}$. What is the probability that at most two requests are received?

$\mathcal{A}$. $\Pr(X \leq 2) = \sum_{k=0}^{2} e^{-3.2} \frac{3.2^k}{k!}$.

Binomial vs. Poisson

Both distributions count the number of occurrences of an event, but they arise in different settings.

	Binomial	**Poisson**
What is fixed?	Number of trials n	Size of interval
What is counted?	Number of successes	Number of occurrences
Parameters	n, p	λ

Example:

- Number of defective items among 100 tested items $\rightarrow$ Binomial.
- Number of defects in one meter of material $\rightarrow$ Poisson.



Poisson Approximation to the Binomial

Suppose $X \sim \text{Binomial}(n, p)$.

When n is large and p is small, the binomial distribution can be approximated by a Poisson distribution with

$$\boxed{\lambda = np}.$$

Thus,

$$\binom{n}{k} p^k (1-p)^{n-k} \approx e^{-np} \frac{(np)^k}{k!}.$$



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Intuition: Many trials + very small probability of success $\rightarrow$ Poisson distribution.





Thank You for your attention!

Any questions?